

VET437 – Hematology of Birds and Reptiles

Lecture Notes

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Introduction

This lecture focuses on the essential knowledge clinicians need for performing a CBC and interpreting hematologic data in birds and reptiles.

Learning Objectives

- Learn how to collect and submit blood for CBC
- Understand basic laboratory methods for performing a CBC
- Understand the limitations of these methods
- Recognize the different types of blood cells and common hemoparasites
- Learn to interpret the most common abnormalities in birds and reptiles

Blood Collection

Timely blood collection after the patient arrives in your care is critical to reduce stress-induced changes in hematologic values. Sedation, such as intranasal administration of midazolam and butorphanol (2 mg/kg each), is often necessary for safe and effective sampling. In birds, venipuncture is commonly performed at the right jugular vein (because it is larger than the left), the ulnar vein, or the medial metatarsal vein.

Species-specific adaptations are necessary; for example, pigeons have a subcutaneous venous plexus in the neck area near the jugular vein, and other sites are recommended for blood collection in this species. Owls have dense feathers on their legs, which conceal the blood vessels; therefore, collecting blood from the jugulars or ulnar veins is better, as they can be visualized. Penguins have very dense plumage and lack the featherless track that birds of other species have over their jugulars; therefore, blood collection is typically done by attempting to feel the jugular, using a blind technique, or by using alternative sites.

Blood Collection in Reptiles

Venipuncture sites in reptiles are more variable by species than in birds, and include the right jugular vein, ventral tail vein, brachial vein/sinus, heart, and dorsal coccygeal vein.

A potential challenge in reptiles is the contamination of blood samples with lymph due to their higher lymph volume compared to blood. This can result in significant changes, particularly on the CBC and to a lesser extent, the biochemistry profile. On a complete blood count, most cell types, except lymphocytes, will be decreased, with the most noticeable change being in the red blood cells and hematocrit. On the biochemistry profile, most parameters remain unchanged; however, the total protein and potassium levels can be reduced.

Lymph contamination is more prone to happen when blood is collected from certain sites. In chelonians, collection from the jugular vein is considered one of the best sites to

avoid blood contamination, followed by the brachial vein. In contrast, collection from the dorsal coccygeal vein and subcarapacial plexus are most prone to lymph contamination.

Sampling Volumes and Materials

The safe maximum blood collection volume is approximately 1% of the animal's body weight, or 0.5% in sick patients. For very small birds, 0.3 mL (1% of a 30g patient) is sufficient for a packed cell volume (PCV), a bloodsmear for a white blood cell estimate and morphology review, and a basic biochemistry profile.

Use the smallest appropriate gauge needle (28G–22G) for the size of the blood vessel. I recommend avoiding pre-heparinized syringes to prevent dilution and anticoagulant misratios if you do not manage to obtain as much blood as you were hoping for when calculating the amount of heparin to preload in your syringe.

Sample Processing and Anticoagulants

Whole blood is used for CBCs, while plasma (anticoagulated) or serum (after clotting) is typically used for biochemistry profiles. However, certain point-of-care and in-clinic analyzers (e.g., Vetscan, blood gas machines, etc.) use heparinized whole blood for some biochemistry analytes.

For CBCs EDTA is the preferred anticoagulant by default due to less thrombocyte (+/- leukocyte) clumping than heparin and better staining quality on the bloodsmear. However, in some species, EDTA causes cell lysis and heparin has to be used. This includes:

- Birds: Corvidae, cranes, ostriches, turacos
- Reptiles: Turtles, tortoises, skinks

Always check if EDTA is safe to use when working with a species you are not familiar with!

Heparin is also the anticoagulant of choice when a limited blood volume is available (e.g., small patients, difficult venipuncture), as one tube of blood can be used for the CBC and biochemistry profile. Typically, blood is collected from the tube for the blood smear, cell count, and to fill a microhematocrit tube (for PCV, total solids, +/- fibrinogen, for which two tubes are needed); and all of the leftover blood can be used for the biochemistry. This is more efficient than filling a tube with blood for a CBC and a separate tube with blood for the biochemistry.

Complete Blood Count (CBC)

1. Blood smears:

Blood smears should be made immediately after blood collection if processing will be delayed (for example, if the sample is shipped to an outside lab), as cell morphology, particularly in leukocytes, can degrade over time, or if the sample is shaken or exposed to temperature extremes during transportation. Air-dried and unstained smears are stable for a long time (weeks to months) at room temperature, so you do not need to fix or stain the smears immediately.

2. Microhematocrit tube:

Filling one or two microhematocrit tubes with blood followed by centrifugation provides a few useful parts of the CBC.

PCV: A PCV is obtained by centrifugation of blood in a microhematocrit tube and measuring the percentage of blood volume composed of red blood cells. In birds, the normal PCV ranges from 38% to 60%; values below 30% often indicate anemia. In reptiles, the normal PCV is between 25% and 35%.

Plasma color: Plasma color is typically influenced by dietary pigments rather than pathology. Many species have yellow-colored plasma, and this should not be interpreted as icterus, as seen in mammals. Additionally, lipemia is common in postprandial or reproductive individuals, especially during vitellogenesis in females. Reptile plasma color is more variable by species than in birds, and some species have green plasma. This includes several species of skinks that have dark green plasma due to high biliverdin levels, which is normal for them.

Total Solids: Once the PCV has been measured, the tube is snapped, and total solids are measured on the plasma, using a refractometer. While total solids are a decent estimate of total proteins in mammals, this is not the case in birds and reptiles. This is in part because they normally have much lower protein levels, than mammals, and potentially higher concentrations of certain electrolytes, and especially lipids (vitellogenesis) which can affect total solids. It is best to go by the total protein concentrations on the biochemistry panel.

Fibrinogen (by heat precipitation): A crude measurement of fibrinogen can be done by comparing the total solids before and after heating the blood. The difference is due to the precipitation of fibrinogen in the heated sample. Since fibrinogen is an acute-phase protein, it is primarily used as a marker of inflammation.

3. Hemoglobin:

Hemoglobin concentration is measured after lysis (+/- centrifugation to pellet nuclei) of the red blood cells in the sample. This can be measured through automated analyzers or simpler benchtop spectrophotometric devices. Once we have the PCV (or hematocrit: which is equivalent to the PCV but calculated by automated analyzers), the hemoglobin concentration and the red blood cell count, all red blood cell indices, MCV, MCH and MCHC can be calculated.

Cell Counts and Estimation

1. Automated hematology analyzers:

Automated hematology analyzers are unreliable for birds and reptiles because they are designed for humans and have been adapted to certain domestic mammal species, with characteristics that do not apply to non-mammals. For example, the machines can reliably separate nucleated cells (leukocytes) from non-nucleated cells (red blood cells and platelets). Then they separate the platelets and red blood cells mostly based on their size, and the different types of leukocytes based on their size and either their internal complexity or myeloperoxidase (MPO) content, depending on the analyzer. MPO present in neutrophils, monocytes and eosinophils but not lymphocytes and basophils. These techniques don't work in non-mammals for multiple reasons, the main one being that all cells are nucleated. Also, bird heterophils (neutrophil equivalent) do not contain myeloperoxidase.

This doesn't mean that automated analyzers are completely useless: they can be used to measure the hemoglobin concentration and generate a red blood cell count. This is because all cells are counted (since the machine cannot reliably differentiate), but the red

blood cells significantly outnumber the leukocytes and thrombocytes so the total cell count is a good approximation of the red blood cell count.

2. Manual Counts with a hemocytometer:

Leukocyte counts must be performed manually, and a count using a hemocytometer is considered the gold standard for this. A hemocytometer is a glass slide designed to hold a specific volume of fluid. Cells are counted within this known fluid volume and a concentration of cells can be obtained. While cell counts are sometimes performed on unstained fluid; for example, this is how RBC and WBC counts are typically performed on cerebrospinal fluid (CSF); non-mammalian blood is often mixed with a special stain first. The stain serves two functions: they dilute the blood to a manageable cell concentration and makes it easier to identify or differentiate the various cell types. The most commonly used stains are "Natt and Herrick's" and "Phloxine B", which you will get to practice working with in the clin path lab!

Natt and Herrick's: Natt and Herrick's (NH) is a purple stain that dyes the nuclei of all cells, as well as the cytoplasm of leukocytes. It's usually used to count leukocytes, but is also marketed for red blood cell counts, and, in theory, could be used to count thrombocytes, although it's not typically used for this purpose. The advantage of NH is that it can stain cells in any species. The disadvantage is that it can be very challenging to differentiate between thrombocytes and lymphocytes (a challenge that is more pronounced in some species than others) in the hemocytometer, which may call into question the validity of the results.

Phloxine B: Phloxine is a bright coral colored stain that stains heterophils and eosinophils. The counts, therefore, are used to calculate the concentration of these two cell types. The total leukocyte count is indirectly calculated after a leukocyte differential is performed on the smear, which provides the percentage of each type of leukocyte. The advantage of this technique is that it tends to be easier and faster to perform (no agonizing over what is and isn't a thrombocyte). However, it appears to work best in birds but seems somewhat less reliable in other species. Some labs use it for birds and reptiles, while others favor NH for reptiles. Additionally, it is unclear whether phloxine stains blue eosinophils (although parrots rarely exhibit eosinophilias so this may be a moot point); and the stain may be fainter in patients with markedly left-shifted and/or toxic heterophils.

Estimates from a smear: Leukocyte counts can be estimated from blood smears by counting the leukocytes in 10 fields of view, using the 40× objective, and multiplying the average count per field by 1600 (or 2500 if using a 50x objective). This is less precise than a hemocytometer count because it is affected by the blood smearing technique, and the location where the count is performed on the smear (some areas within one smear are thicker than others). Additionally, multiple formulas are found throughout the literature, and the most appropriate formula to use should ideally be tailored to the specific microscope being used. Ideally, the correct formula should be determined for each microscope by comparing the estimated leukocyte count with the counts obtained using a gold standard; however, most people don't bother. Decent reproducibility can be achieved if the same microscope and smearing technique are always used.

Biological and Laboratory Variability

Biological and laboratory variability are of greater importance in non-mammals than in mammals, particularly in reptiles, which are ectotherms and therefore have a

metabolism that is significantly influenced by environmental factors, such as temperature, season, and hibernation, which complicates the establishment of universally applicable reference intervals.

Variability also arises from biological differences between species and sexes, particularly in females as they prepare to lay eggs. In addition, manual techniques are more prone to variation due to a combination of inherent imprecision, resulting from counting low numbers of cells (dozens to hundreds when counting manually vs. 10,000 for mammals in an automated analyzer), and being more susceptible to human error. The main consequence of this is that when monitoring CBC values to determine if a patient is improving or worsening, you need to keep in mind that relatively mild changes (eg: 10-15%) might be due to technique imprecision and doesn't necessarily mean that the numbers have actually changed within the patient.

Erythrocytes

Birds have a short erythrocyte lifespan (35–45 days) while reptiles tend to have longer RBC lifespans (300–600+ days). Because of the fast turnover of erythrocytes, birds usually have polychromatophils in circulation (with higher numbers with regeneration), while reptiles don't and typically show polychromasia only with a regenerative response. Both birds and reptiles that exhibit strong regenerative responses can have relatively early erythroid precursors in circulation, compared to what is observed in mammals. Additionally, reptiles may display mitotic figures, binucleation, or unusual morphology in red blood cells during regeneration.

Polychromatophils are typically quantified on the blood smear and usually reported as the number (or more likely a range) per field of view. I find that there is some subjectivity to this, as there is a continuum in the maturity level of red blood cells on the smear, and two different observers may not classify cells as polychromatophils in the same way (i.e., how blue does the cytoplasm have to be to be counted as a polychromatophil?). In addition, as for the estimated leukocyte count on the smear, the number will vary with the thickness of the blood smear, and therefore, where you do the counting on the smear.

Reticulocyte counts tend to provide a more accurate count because they are expressed as a percentage of red blood cells, and if the RBC count is available, this number is converted to an absolute count (number of cells/ μ L). Reticulocytes and polychromatophils are both immature red blood cells; however, reticulocytes are identified using special stains (e.g., special fluorescent stains used by hematology analyzers or new methylene blue used for manual counting) whereas polychromatophils are identified using "regular" stains, such as Diffquick, Wright's Giemsa etc... Reticulocytes are identified by the presence and number of dark blue spots in their cytoplasm, which corresponds to ribosomal RNA. In birds, many red blood cells have some of these dark dots visible, in variable numbers, however, only certain types, ring reticulocytes, appear equivalent to polychromatophils. Ring reticulocytes have dots arranged around 50% or more of the nuclear periphery. Ring reticulocytes are poorly characterized in reptiles, so it may not be worth requesting manual reticulocyte counts in reptiles at this time, except perhaps if it is primarily used to compare multiple samples within an individual to look for general trends.

The interpretation of erythrograms and anemias is similar in birds, reptiles, and mammals. Some differences include that anemia secondary to inflammatory disease can occur more rapidly and be more severe, particularly in birds, due to the shorter lifespan of RBCs. In addition, remember that a low PCV can indicate lymph contamination in reptiles,

rather than true anemia. Regeneration, particularly in birds, can occur faster. Immune-mediated hemolytic anemias are uncommon and won't cause hyperbilirubinemia.

Thrombocytes

Thrombocytes are nucleated cells responsible for hemostasis, and are therefore considered equivalent to platelets, however, they have slightly different functions, for example, they are capable of phagocytosis.

In practice, thrombocyte counts are not scrutinized as much as platelet counts in mammals. This may be because they are often clumped, particularly in heparinized samples, which makes obtaining an accurate count challenging. Additionally, diseases causing thrombocytopenia may be uncommon (or possibly underdiagnosed?).

They are often confused with lymphocytes on blood smears and are most problematic when they cause an overestimation of the lymphocyte (and therefore also leukocyte) count.

Leukocytes and Differentials

Leukocyte differentials must be performed manually by counting 200 cells on a smear. The predominant leukocyte type in health varies by species—heterophils predominate in psittacines, while lymphocytes are more common in other birds.

Lymphocytes: They have the same morphology as in mammals. Most are small, but intermediate and large reactive lymphocytes can be seen with inflammation/antigenic stimulation. Plasma cells can occasionally be seen in circulation with inflammation, particularly in reptiles. Lymphopenia can occur with stress leukograms. It is not clear how high the lymphocyte count can go in birds and reptiles, due to antigenic stimulation or infectious diseases and I've heard anecdotes from colleagues of cases with very high counts ($>40,000/\text{ul}$) that have gone down on their own. It is therefore difficult to give you a cut off beyond which we can differentiate between antigenic stimulation vs. neoplasia. However, chronic lymphocytic leukemia is not all that uncommon, and lymphoma and all types of acute hemic neoplasia (leukemia) can also occur, and should be the top differential in progressive or persistent marked lymphocytosis (especially if $>20,000/\text{ul}$).

Monocytes: They have the same morphology as in mammals, however, in reptiles there is a cell type that is like monocytes but with fine pink granules in the cytoplasm. These are called azurophils. Some species may have both cell types or only one. The function of azurophils may be closer to mammal neutrophils in some species (snakes and lizards) in terms of the speed at which they arrive a site of inflammation, but could be closer to monocytes (slower to arrive) in others. Also, it more common to see monocytes that are activated (macrophages) and show signs of phagocytosis in circulation, secondary to inflammation in reptiles. Melanomacrophages are a cell type found in certain organs of reptiles and are occasionally observed in circulation, particularly during inflammation. While the monocyte counts tend to not be very high in mammals, even with significant chronic inflammation, in birds, the monocyte counts seen with certain types of chronic infectious diseases, often accompanied by heterophilia, can be very high ($>60,000/\text{ul}$). This is most often seen with Mycobacteriosis, Aspergillosis and Chlamydiosis – remember these differentials when you have a very high leukocyte counts made up of monocytes and/or heterophils in birds.

Heterophils: They are the equivalent of neutrophils, even though they are not exactly the same (eg: the enzymes and other content of granules are different). Heterophils have pink granules filling their cytoplasm. These are usually rod-shaped in birds, sometimes with a visible refractile structure in the center of the granules. In reptiles, the shape of the granules is more variable, ranging from rod-shaped to round, and sometimes they seem to melt or blur together. Heterophils increase in response to acute and ongoing inflammation, as well as during a stress leukogram. Left-shifting is interpreted the same as in mammals. The toxic changes that occur in mammals (cytoplasmic basophilia, Doehle bodies and vacuolation) also occur in birds and reptiles, but this is often hidden by the granules. Therefore, the best way to identify toxic changes is by the decreased number of pink granules in the cytoplasm. Since immature heterophils also have fewer pink granules, it can be challenging to determine whether heterophils have few granules due to left shifting or toxicity, or both. Heterophils in birds are segmented, however, the heterophils in some reptiles have round nuclei. Therefore, detecting left shifting (aka: immature heterophils) is easier in some species (the ones with normally segmented nuclei) than others (those with normally round nuclei).

Eosinophils: Eosinophils have a morphology that is more variable than any other cell type (see images in lecture slides). In birds, psittacines often have round, pale blue granules, while other species have pink ones. Differentiating eosinophils from heterophils can be challenging, but eosinophils have brighter pink granules, their granules are often round (vs. rod-shaped), they tend to be slightly larger, and have slightly more basophilic cytoplasm (not always visible between the granules). In reptiles, eosinophil granules can also vary in color from blue to pink and exhibit significant variations in size and number.

Eosinophilia appears more common in some species than others (eg, raptors and chelonians), and does not appear to be as specific to parasitic and allergic diseases in these species as in mammals. It is likely a more general marker of inflammation. It is not uncommon for eosinophils (or basophils) to be the predominant leukocyte type with inflammation in some species, particularly chelonians.

Basophils: They are typically easy to identify due to their dark purple granules. However, they are commonly lysed, so you may only be able to see a bare nucleus (round in these species) surrounded by a halo of floating purple granules when evaluating a blood smear. I often get asked how one differentiates a basophil from mast cells in these species, if basophils have round nuclei, and the answer is: I don't know. While mast cell tumors can occur in reptiles, I've never had a case that had mast cells in circulation (that I know of) that I could use for comparison. As in eosinophils, they appear to be part of more generalized inflammatory responses, as opposed to being specific to certain diseases (eg: parasites).

Common Hemoparasites

Hemoparasites are a fairly common finding in wild birds and reptiles, or those housed outdoors. Leukocytozoon, Hemoproteus and Plasmodium are most common.

Leukocytozoon are large hemoparasites which can be round or spindle shaped. They are actually parasites (micro- or macro-gametes) within a nucleated cell that has been deformed beyond recognition. There have been debates about what the cell of origin is, with leukocytes and erythrocytes being the top contenders. A few years ago a study showed that, at least in some cases, the parasites were in fact within thrombocytes. Leukocytozoon usually does not cause clinical signs and is often an incidental finding, particularly in

raptors. When increased in number in a sick bird it is often assumed that their number increased secondary to the fact that the birds were immunocompromised. With this said, the clinical signs secondary to Leukocytozoon is dependent on the species and clinical signs (and occasionally death) can occur in geese, ducks, chickens and turkeys.

There is enough morphological overlap between Hemoproteus and Plasmodium that PCR is recommended to differentiate between these two. Although, in general, it is said that if the parasite displaces the nucleus of the red blood cell it's more likely to be Plasmodium, vs. if it wraps around without moving it. These also often cause no, or minimal disease, in birds, particularly those who have evolved with these parasites (eg: local wildlife). However, in other species, Plasmodium in particular, can cause significant disease and death (eg: in arctic species like penguins).

Resources

If you need extra information and images to help you with avian and reptile hematology you can try:

1. Clinical Pathology sections of exotic medicine books. For example:
 - [Exotic Animal Formulary](#) – for reference intervals
 - [Reptile Medicine and Surgery](#) – for interpretation and reference intervals
 - [Current Therapy in Avian Medicine and Surgery](#) – for interpretation
2. Species-specific chapters within general clinical pathology books, or dedicated exotics clinical pathology books:
 - [Exotic Animal Hematology and Cytology](#)
 - [Schalm's Veterinary Hematology](#)
 - [Veterinary Cytology](#)
3. Scientific Journals
 - [Veterinary Clinical Pathology](#)
 - [Journal of Avian Medicine and Surgery](#)
 - [Journal of Zoo and Wildlife Medicine](#)